

Pawan Dilip Chaudhari

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SUMMARY

Aspiring AI Engineer with a strong research background, featuring three peer-reviewed publications in major IEEE conferences regarding Cybersecurity and Healthcare AI. Developed an Industrial ABSA Engine and deployed Llama 3 models alongside dual-stream Deepfake analyzers and IoT systems. Proficient in Python, TensorFlow, PyTorch, Docker, FastAPI, and React.js, bridging the gap between advanced Machine Learning models, DevOps infrastructure, and functional Web Development. Professionally certified by global leaders including IBM, Google, and **The University of Edinburgh** in Data Science and Cybersecurity. Committed to engineering explainable AI solutions that solve complex real-world challenges through data-driven innovation.

EDUCATION

Datta Meghe Institute of Higher Education and Research, WARDHA **2023 - Present**
Bachelor of Technology **Artificial Intelligence and Data Science**

Full-time | SGPA: 6.86/10 5th sem | CGPA 6.30/10
Studying at the Faculty of Engineering & Technology (FEAT), DMIHER, a NAAC A++ accredited institution with a CGPA of 3.78/4, recognized for excellence in research and innovation.

Harikamal Science Junior College Kharadgaon, Ner **2021 - 2023**
Higher Secondary Certificate (12th) **General Science (PCMB)**

Maharashtra State Board, Amravati Division | Percentage: 61.83%

Elephant I.F English Medium High School, Ner **2020 - 2021**
Secondary School Certificate (10th)

Maharashtra State Board, Amravati Division | Percentage: 91.20%

PROFESSIONAL EXPERIENCE

AI / ML INTERN **6 Months**

Ganishka Technology (A Division of Ganishka Enterprises)

Key Responsibilities & Achievements:

- **LLM Engineering (Llama 3 - 8B):** Implemented and executed Meta's open-source large language model locally with optimized GPU acceleration routines for targeted RAG interfaces and customized Q&A fine-tuning.
- **Multidomain ABSA Engine:** Formulated an industrial Aspect-Based Sentiment Analysis (ABSA) engine processing 58,332+ deduplicated lines across 1,135 subsets, achieving **84.2% SVM accuracy** and **81.9% Logistic Regression accuracy**.
- **Infrastructure & DevOps:** Sealed runtime script dependencies inside Docker containers, integrated decoupled FastAPI endpoints with background MySQL setups, and conducted multi-layer data deduplication filters to eliminate statistical leakage.

Skills Gained: Llama 3 (8B), RAG, Docker, FastAPI, Python, Pandas, NumPy, Scikit-Learn, SVM, MySQL, TF-IDF Vectors, Git.

WEB DEVELOPER **April 2025 - June 2025**

Training & Internship, Cosmos Digital X ILA

Key Responsibilities & Achievements:

- **Front-End:** Utilized React.js to build user-friendly, responsive UI components for company projects.
- **Back-End:** Developed backend functionality to support upcoming company applications and integrated APIs for dynamic data handling.
- **Data Visualization:** Created analytical dashboard pages designed to visualize client-side trends and internal metrics.
- **Professional Growth:** Demonstrated initiative and commitment to deepening technical skills through rigorous professional training.

Skills Gained: React.js, JavaScript, HTML/CSS, Backend Development, API Integration, Data Visualization, Analytical Dashboard Design, Team Collaboration, and Performance Optimization.

PROJECTS

Artificial Intelligence for Deepfake Identification: A Dual Analyzer for Audio and Video **Dec 2025 - Feb 2026**

- Engineered a dual-stream AI analyzer to identify manipulated media across both video and audio channels, achieving **94.2% accuracy** across diverse datasets.
- Implemented an audio pre-processing pipeline using Librosa to extract MFCCs and spectral features, reducing feature extraction latency by **40%**.
- Developed a frame-level spatial-temporal analysis framework using CNN-RNN architectures to detect facial inconsistencies and synthetic audio artifacts.
- Tools & Dataset: TensorFlow, Keras, OpenCV, Librosa; Trained on FaceForensics++ and DFDC datasets.

Automated Glaucoma Identification & Clinical Decision Support System

July 2025 - Oct 2025

- Engineered a deep learning diagnostic system using ResNet-50 and U-Net architectures to automate Glaucoma identification with **96.5% sensitivity**.
- Integrated Grad-CAM visualization to overlay class-discriminative heatmaps on fundus images, enabling transparent and explainable AI decision-making.
- Pre-processed high-resolution clinical images using stratified data splitting and weighted loss functions to optimize Cup-to-Disc Ratio (CDR) calculations.
- Tools & Dataset: PyTorch, Matplotlib, Flask, CUDA; Trained on ACRIMA and Drishti-GS datasets.

Development of a Smart Car Parking System

Jun 2024 - Aug 2024

- Architected an end-to-end IoT ecosystem using ESP32/Arduino and ultrasonic sensors, achieving **99% real-time occupancy detection accuracy**.
- Integrated hardware sensors with a Firebase real-time database.
- Optimized allocation algorithms to maximize space efficiency, resulting in a **40% reduction** in average parking search time for users.
- Hardware Arduino IDE.

TECHNICAL SKILLS

Programming Languages: Python, Embedded C++, JavaScript, SQL.

Machine Learning & LLMs: Llama 3 (8B), RAG (Retrieval-Augmented Generation), Deep Learning, Transfer Learning, CNN, RNN, SVM, Logistic Regression, Grad-CAM, Aspect-Based Sentiment Analysis (ABSA).

Frameworks & Libraries: TensorFlow, Keras, PyTorch, FastAPI, React.js, Scikit-Learn, OpenCV, Pandas, NumPy, Librosa.

Data Visualization: Tableau, Power BI, Matplotlib, Seaborn, Plotly.

DevOps & Tools: Docker, Git, Kaggle, Jupyter Notebook, Google Colab, VS Code, Arduino IDE.

Databases: MySQL, MongoDB, Firebase.

PROFESSIONAL SKILLS

Problem Solving: Ability to develop innovative solutions for complex technical challenges, such as multi-modal AI, LLM deployment, and IoT optimization.

Research & Development: Experienced in implementing cutting-edge technologies from research and documenting findings for technical audiences.

Project Management: Skilled in planning and delivering technical projects on time while managing resources effectively.

RESEARCH PUBLICATIONS

Development of a Smart Car Parking System - Engineered a technology-driven IoT solution featuring real-time occupancy detection using ESP32 and ultrasonic sensors to optimize urban parking efficiency. View on IEEE Xplore	IDICAIEI, 2024
Deep Learning Approach and its Application in the Cybersecurity Domain - Investigated advanced neural network architectures to enhance threat detection and security protocols within digital environments. View on IEEE Xplore	ICMLAS, 2025
Advanced Lung Cancer Diagnosis and Treatment with Artificial Intelligence - Explored AI applications and future directions to improve diagnostic accuracy and treatment methods in clinical oncology. View on IEEE Xplore	ICEARS, 2025

ADDITIONAL INFORMATION

Languages: English, Hindi, Marathi.

Certifications:

- Foundations of Cybersecurity** - Google (Coursera) | Dec 2024
- Python for Data Science, AI and Development** - IBM (Coursera) | Dec 2024
- Generative AI: Enhance your Data Analytics Career** - IBM (Coursera) | Dec 2024
- Data Science in Stratified Healthcare and Precision Medicine** - The University of Edinburgh (Coursera) | Nov 2024